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Lesson Notes Part I

MODULE 1 | LESSON 1 | PART I

SAMPLE MOMENTS AND PORTFOLIO PERFORMANCE - PART I

Reading Time	60 minutes
Prior Knowledge	Basic statistics and a working knowledge of basic portfolio theory
Keywords	Return and variance of a portfolio, Beta of a portfolio and relevant types of risk, Performance appraisal (Sharpe and Treynor ratios, Sortino ratio, information and appraisal ratio, skewness, and kurtosis)

This lesson introduces financial portfolios and discusses how to compute certain moments like mean and standard deviation as well as a measure of systematic risk, like the beta. Then, after a discussion of risk, we look at how to assess the performance of portfolios in regards to particular risks. At this stage, the traditional indices used for performance appraisal are introduced, like the Sharpe and Treynor ratios. Furthermore, we discuss how volatility is bad only when it is on the downside and not on the upside, and we also consider other measures of appraisal. Finally, we discuss the implications of skewness and kurtosis and what impact they could have on performance if assessed without regard to these statistics.

1. Sample Moments and Relevant Vector Notation

An investment portfolio is defined as a collection of a variety of assets (stocks, bonds, money market accounts, exchange-traded funds, closed-end funds, real estate, private investments, etc.). Some of these assets were mentioned individually in previous courses. This lesson is not about the specifics of any individual assets; instead, it is focused on an aggregate class of investments. We will learn how to describe portfolios' returns and their risk potential. In this module and the following modules, we will see how to create optimal portfolios and some of the nuances and weaknesses that come with it.

To work effectively with portfolios of financial assets, we must begin with a formal definition. A portfolio consisting of n assets, will be represented by a vector of weights $w = (w_1, w_2, \ldots, w_n)$.

The weights are expressed as the percentage of total wealth allocated to each particular asset. If

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all weights are non-negative, and they sum up to 1. Whenever short-selling is allowed, it will be mentioned explicitly.

In working with portfolios, depending on the specific applications, it might be necessary to use the language of vectors and matrices. When mathematically convenient, we will use this language to compute certain quantities for our portfolios.

To this purpose, we introduce some basic notation. If x is an $n \times 1$ vector, we will write it as the column vector:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

and its transpose x^T will denote the $1 \times n$ row vector

$$x^T = (x_1 \quad x_2 \quad \dots \quad x_n)$$

Furthermore, by the $n \times m$ matrix X, we mean that

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1m} \\ x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nm} \end{pmatrix}$$

Then, if $r^T = (r_1, r_2, \dots, r_n)$ represents the vector of returns for each of the n assets in a portfolio, and

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_{nn} \end{pmatrix}$$

is the variance covariance matrix of the n-assets, we can write the return for the portfolio as

$$r_p = \sum_{i=1}^{n} w_i r_i = w^T r \text{ and } \sigma_p^2 = \sum_{i=i}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij} = w^T \Sigma w.$$

2. Portfolio Beta

The beta of a security or a portfolio is a measure of the volatility or systematic risk of this security or portfolio compared to the market as whole. The required rate of return on an asset can be related to its systematic and non-systematic risk by writing the capital asset pricing model (CAPM) for the asset a as:

$$E[r_a] = r_{RF} + \beta(E[r_m] - r_{RF})$$

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where r_m represents the market return, r_{RF} is the risk-free rate, which we assume to be constant over the reference period, and ϵ is an error term such that $E[\epsilon] = 0$ represents the asset's risk not related to market conditions. This model can be re-written as:

$$r_a - r_{RF} = \beta(r_m - r_{RF}) + \epsilon$$

Then, a simple application of results from linear regression yields

$$\beta = \frac{cov(r_a - r_{RF}, r_m - r_{RF})}{var(r_m - r_{RF})},$$

where var and cov are short forms for variance and covariance, respectively.

The beta of a portfolio, say β_p, can be easily obtained from the betas of the individual assets that make up the portfolio. That is, if we represent the beta of each of the n assets as a column vector, i.e.:

$$\beta = (\beta_1\ \beta_2\ \dots\ \beta_n)$$

and w is the vector of the assets' weights in the portfolio, then the formula for β_p is simply:

$$\beta_p = w^T\beta = \sum_{i=1}^{n} w_i\beta_i.$$

To check that the beta of a portfolio indeed has a linear structure, it is sufficient to recognize that, according to the definition of beta we gave above, it follows that

$$\hat{\beta}_p = \frac{cov(r_p - r_{RF}, r_m - r_{RF})}{var(r_m - r_{RF})}.$$

However, we saw earlier that $r_p = \sum_{i=1}^{n} w_i r_i$ and that $\sum_{i=1}^{n} w_i = 1$ and therefore:

$$\beta_p = \frac{cov(\sum_{i=1}^{n} w_i(r_i - r_{RF}), r_m - r_{RF})}{var(r_m - r_{RF})}$$

$$\beta_p = \frac{\sum_{i=1}^{n} cov(w_i(r_i - r_{RF}, r_m - r_{RF}))}{var(r_m - r_{RF})}$$

$$\beta_p = \sum_{i=1}^{n} w_i \frac{cov(r_i - r_{RF}, r_m - r_{RF})}{var(r_m - r_{RF})}$$

$$\beta_p = \sum_{i=1}^{n} w_i\beta_i$$

using the definition of β_i $\textbf{for each}\ i = 1, 2, \dots, n$ as claimed. From this derivation, it should be clear that the beta of a portfolio is a linear combination of the betas of its assets through the li y (bi-linearity, actually) and the other properties of the covariance operator.

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portfolio is indicated in the table immediately below:

	Asset 1	Asset 2	Asset 3
beta	0.5	1.6	1.2
$ amount	$250,000	$350,000	$400,000

What is the beta for this portfolio?

The dollar value of our portfolio is

$\$250,000 + \$350,000 + \$400,000 = \$1,000,0000$.

Then, the weights for each of the assets in the portfolio can be computed as

$$w_1 = \frac{250,000}{1,000,000} = 0.25, w_2 = \frac{350,000}{1,000,000} = 0.35, w_3 = \frac{400,000}{1,000,000} = 0.4.$$

Thus, the beta for our portfolio turns out to be

$$\beta_p = \sum_{i=1}^{3} w_i \beta_i = 0.25 \times 0.5 + 0.35 \times 1.6 + 0.4 \times 1.2 = 1.165.$$

You will learn about risk management in a different module, but we can still use this example to illustrate the use of the betas of assets.

Suppose that the portfolio manager is concerned that the overall market is going to be bumpy during the next 12 months and, as such, would like to eliminate the systematic risk in their portfolio.

The S&P 500 Index is currently priced at 1,000 and a futures contract on the S&P 500 index is currently priced at $1,010 and has a beta of 1.05. Each future contract is $250 times the index price.

The manager does not want to sell the assets because they think the assets are good long-term investments.

How should the manager act to reduce their market exposure during this period?

Reducing market exposure is equivalent to reducing the beta. More specifically, if we create a new portfolio with a beta of zero, we would have created a **beta neutral portfolio**. For this, the manager can use the S&P 500 futures and sell short:

$$n = \left(\frac{\beta_T - \beta_p}{\beta_F}\right) \times \frac{MVP}{F}$$

with β_T the new target beta, i.e., zero in our case, MVP the market value of our portfolio, and F the market value of the futures contract on the S&P500. Then,

$$n = \left(\frac{0–1.165}{1.05}\right) \times \frac{1,000,000}{250 \times 1010} = -4.394.$$

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3. Relevant Types of Risk

It is not sensible to talk about portfolio returns without talking about risk because investment decisions involve a trade-off between return and risk. Investors need to be aware of the risk they are assuming and what kind of return they are getting for taking these risks. This can also be useful when deciding which portfolio managers to hire or fire.

One can identify several types of risk:

- **interest rate risk**, caused by changes in the level of interest rates and which generally affect securities inversely, although it affects bonds more directly;
- **market risk**, due to fluctuations in the overall market and to which all securities are exposed, although common stocks are the most sensitive to it;
- **inflation risk**, which affects the purchasing power of investments. This risk is related to interest rate risk as interest rates usually rise with inflation;
- **business risk**, or the risk of doing business in a particular industry;
- **exchange rate risk**, the consequence of currency fluctuations and concerning portfolios that contain global investments since currency fluctuations can affect the variability of returns;
- **liquidity risk**, which is associated in particular with investments that trade in secondary markets. If an investment cannot be bought or sold quickly because it is not liquid, then there could be "price slippage" between when the decision to buy or sell was made and when the transaction actually took place. This is also seen in primary markets when the size of certain orders is large compared to the volume of trades for that particular security; and
- **country risk or political risk** is relevant to investors investing internationally, due to instability at both a political and economic level.

All these categories are part of what we refer to as total risk. However, it is convenient to categorize sources of risk according to their nature, i.e., systematic risk and non-systematic risk. The key "equation" here is

$$\textbf{total risk} = \textbf{systematic risk} + \textbf{non-systematic risk.}$$

Systematic (or market) risk is the risk attributable to broad macro factors and affects all securities. When an investor constructs a well-diversified portfolio, they can eliminate part of the total risk, and what is left is the non-diversifiable portion of the risk, which is known as systematic risk.

Non-systematic (idiosyncratic) risk represents the variability in total returns of an investment that is not related to overall market variability. This type of risk is unique to a particular security and, among the risks listed above, related to liquidity risk, business risk, and country risk.

We have already learned how to "measure" total risk and systematic risk. Next, we will provide a measure of non-systematic risk, which we will be referring to as σ_u. To compute σ_u, one begins with a version of the one-factor model:

$$r_p = E[r_p] + \beta_p(r_m - E[r_m]) + u.$$

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β_p is a coefficient that measures the portfolio's exposure to the market return r_m, and u is the non-systematic component of the portfolio's return.

The model assumes that r_p and r_m are jointly normally distributed and r_m and u are uncorrelated. Then, using basic results for simple linear regression, we find that

$$\beta_p = \frac{cov(r_p, r_m)}{var(r_m)}.$$

In this one-factor model, the two parts on the right-hand side are uncorrelated with one another and therefore:

$$\sigma_p^2 = \beta_p^2 \sigma_m^2 + \sigma_u^2.$$

From the last equation, we can also write.

$$\sigma_u^2 = \sigma_p^2 - \beta^2 \sigma_m^2$$

and using the value for β_p in this equation, we get, using simple algebraic manipulations, that

$$\sigma_u^2 = \sigma_p^2 (1 - corr^2(r_p, r_m))$$

and we see that if a portfolio has a perfect correlation with the market portfolio, then $\sigma_u^2 = 0$, as we should expect.

4. Performance Appraisal: Sharpe and Treynor Ratios

Managers and investors alike try to appraise the performance of their investments to assess whether investment results are satisfactory. Risk-adjusted performance measures are desirable because it is relatively easy to get lucky by assuming a lot of risk. However, this exposes the investors to potentially serious downsides as well whenever things don't work out as hoped. This is even more important when the wealth of an individual or organization is managed by a third party, like a portfolio manager who is allowed to pursue an active management style in deciding the composition of portfolios for clients. It is important for investors or mutual funds that a portfolio manager's active style adds value on a risk-adjusted basis.

We are now going to look at several appraisal measures, the type of risk they address, and some of the issues.

The notorious Sharpe ratio is calculated as the excess return over the risk-free rate divided by the standard deviation:

$$\text{SR}_p = \frac{\bar{r}_p - \bar{r}_{RF}}{\sigma_p}.$$

where $\bar{r}_p$ and $\bar{r}_{RF}$ denote the mean return on the portfolio and the mean risk-free rate during the r[illegible]ce period, respectively, and σ_p represents the volatility of the portfolio during the same p[illegible] Overall, a Sharpe ratio of less than 1 is considered bad; between 1–1.99, it is considered

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The Treynor ratio is similar to the Sharpe ratio, but the denominator only considers systematic risk rather than total risk, i.e.:

$$T_p = \frac{\bar{r}_p - \bar{r}_{RF}}{\beta_p}.$$

When is the Treynor ratio more appropriate than the Sharpe ratio? Since the denominator in the definition of the Treynor ratio is the beta of that portfolio, it is more appropriate when one can assume efficient markets; that is, the Treynor ratio is a suitable form of portfolio appraisal only in those instances where portfolios just have systematic risk.

The **Modigliani and Modigliani measure** (M2, M^2, RAP, Modigliani risk-adjusted performance) allows us to expand the applicability of the Sharpe ratio. The formula for it is:

$$M2_p = \bar{r}_{RF} + \left(\frac{\bar{R}_p - \bar{r}_{rf}}{\sigma_p}\right)\sigma_m = \bar{r}_{RF} + \mathrm{SR}_p\sigma_m.$$

In essence, M2 is a scaled version of the Sharpe ratio and tells us how much a portfolio would have rewarded an investor with the given amount of risk taken in terms of the risk-free rate of return and benchmark portfolio. In other words, it could help us determine how much return the portfolio would have attained if it had the same risk level and benchmark. Note that if $\sigma_p < \sigma_m$, then the expected return on the portfolio would have been magnified while the opposite is true if $\sigma_p > \sigma_m$.

The M2 overcomes the difficulties associated with the Sharpe ratio and the difficulties with comparing Sharpe, Treynor, Sortino, and other ratios among different investments.

Because M2 yields a risk-adjusted performance measure as a percentage return unit, it is very straightforward to interpret. However, one should notice that the M2 measure is derived directly from the Sharpe ratio, and therefore, any portfolio orderings using the M2 measure will be the same as the portfolio ordering using the Sharpe ratio.

These lesson notes are divided into two parts. Please continue reading about the Sortino Ratio next.

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